

Time Series Forecasting

“Forecasting is not about predicting the future, it’s about reducing uncertainty.” -- Paul Saffo

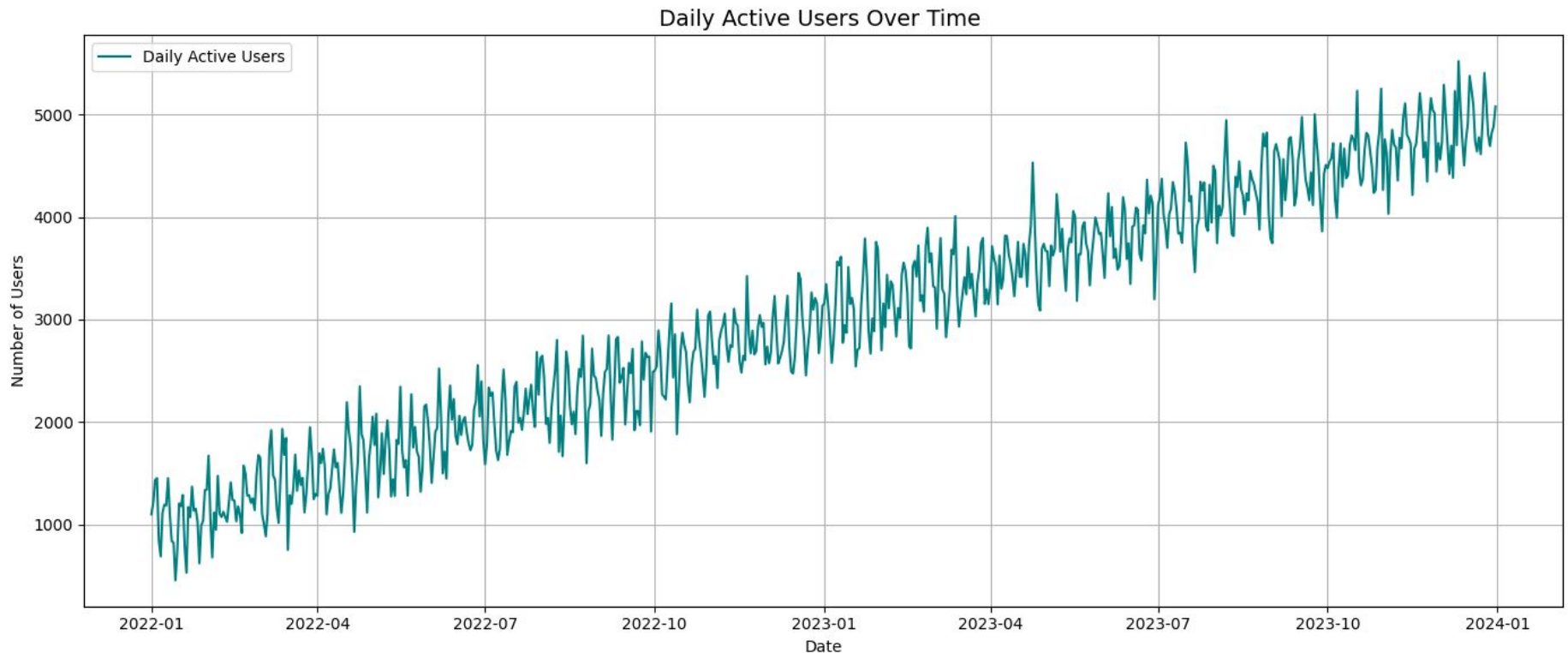
Agenda

- Exploratory Data Analysis
 - Overall Patterns and Trends
 - Seasonality (Recurring Patterns at Fixed Intervals)
 - Cyclicity (Long-Term Non-Fixed Repeating Patterns)
 - Stationarity
 - Autocorrelation (Self-Correlation Over Time)
 - Outliers and Anomalies
 - Distribution of Values
 - Relationships Between Multiple Time Series

Overall Patterns and Trends

Trend Plot

- Trend plots help visualize how a time series behaves over time.
- They reveal long-term increases or decreases in the data

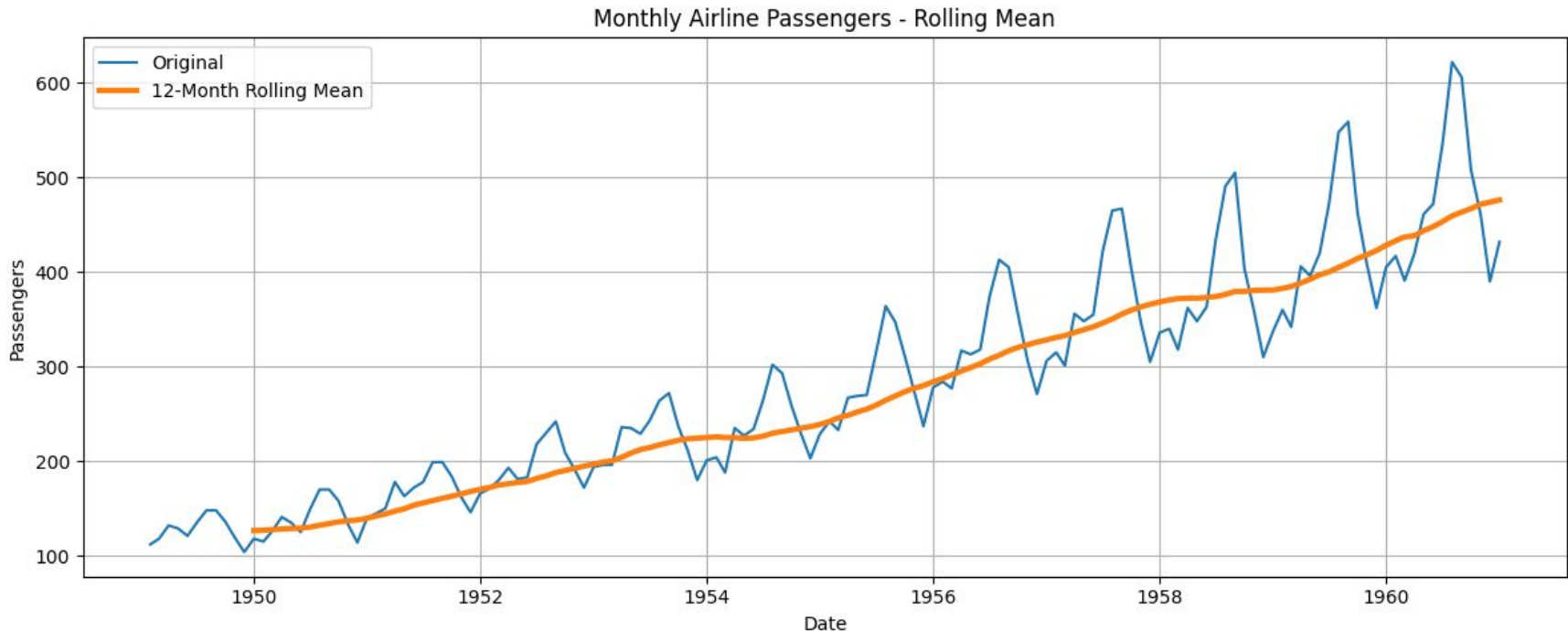


From the plot, we can observe a clear upward trend in daily active users over the 2-year period.

Rolling mean plot

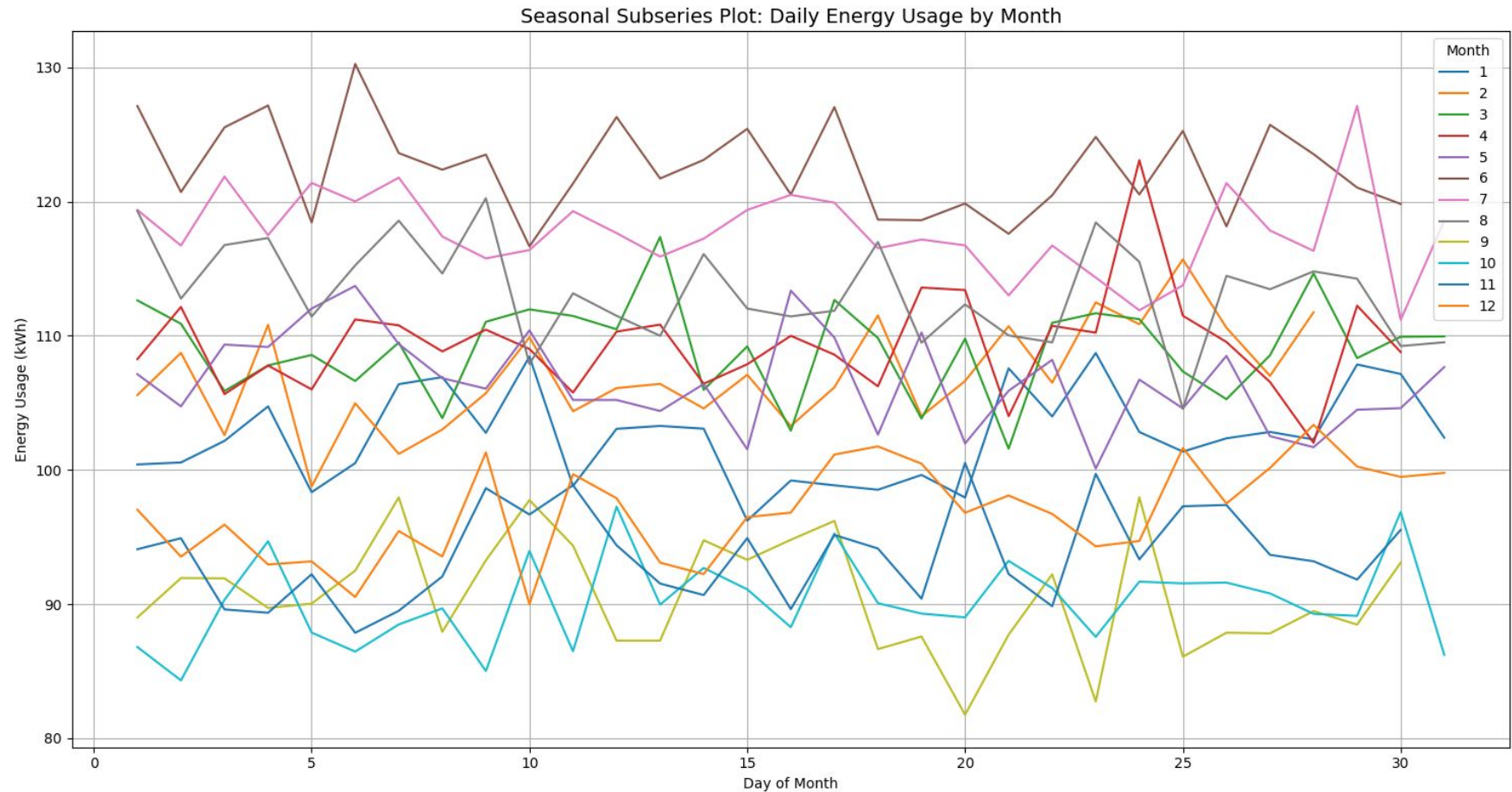
A Rolling Mean Plot (also called a moving average plot) is used to smooth out short-term fluctuations and highlight longer-term trends in time series data.

- Detects **trend direction**.
- Reduces **noise** caused by random fluctuations.
- Helps in understanding **underlying seasonality** and **stationarity**.



- The original time series shows strong seasonality and upward trend.
- The rolling mean smooths out short-term fluctuations and clearly reveals the increasing trend.

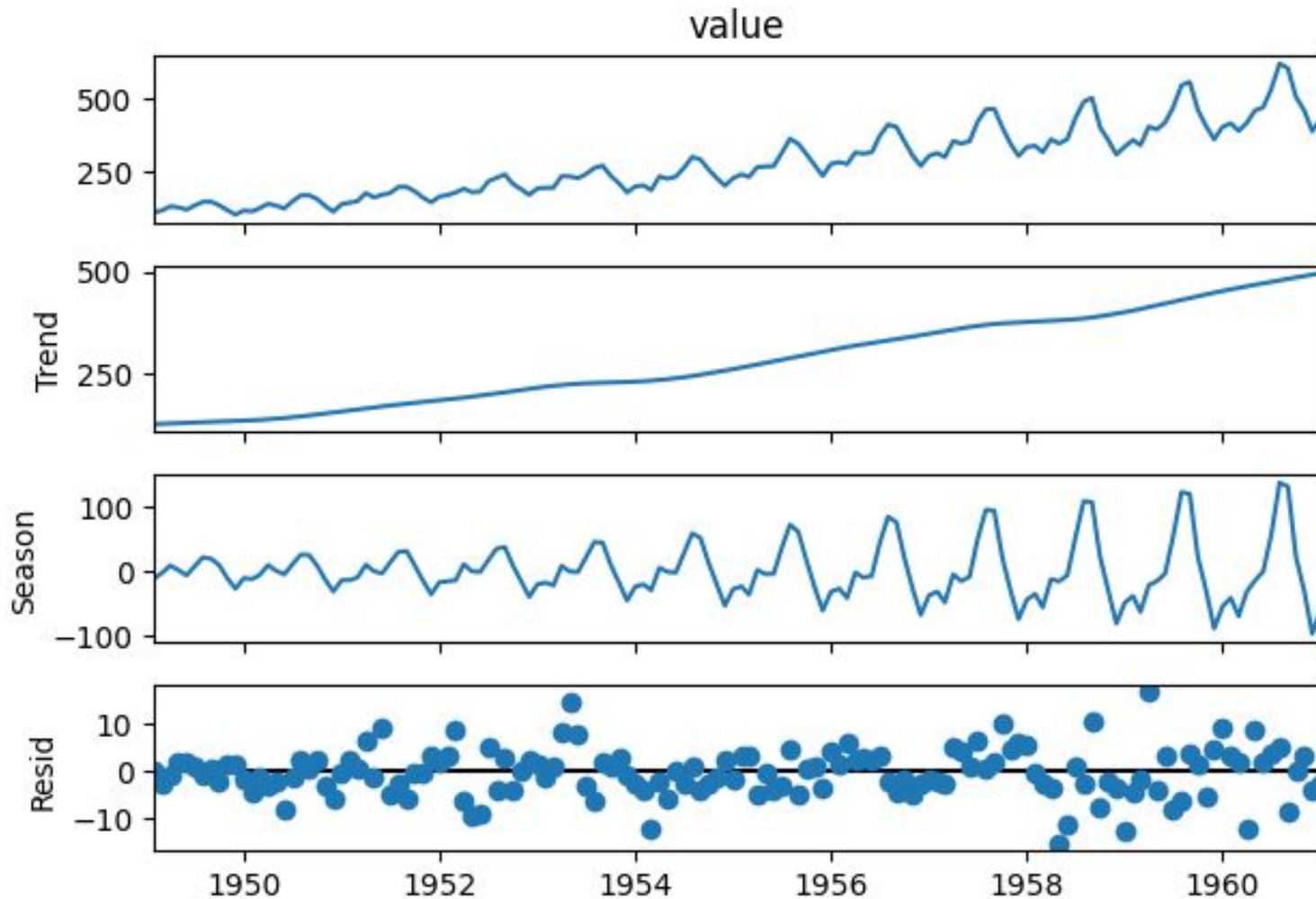
Multiple line plot



higher usage in summer months (June to August) due to air conditioning

Seasonal Decomposition

STL Decomposition - Airline Passengers

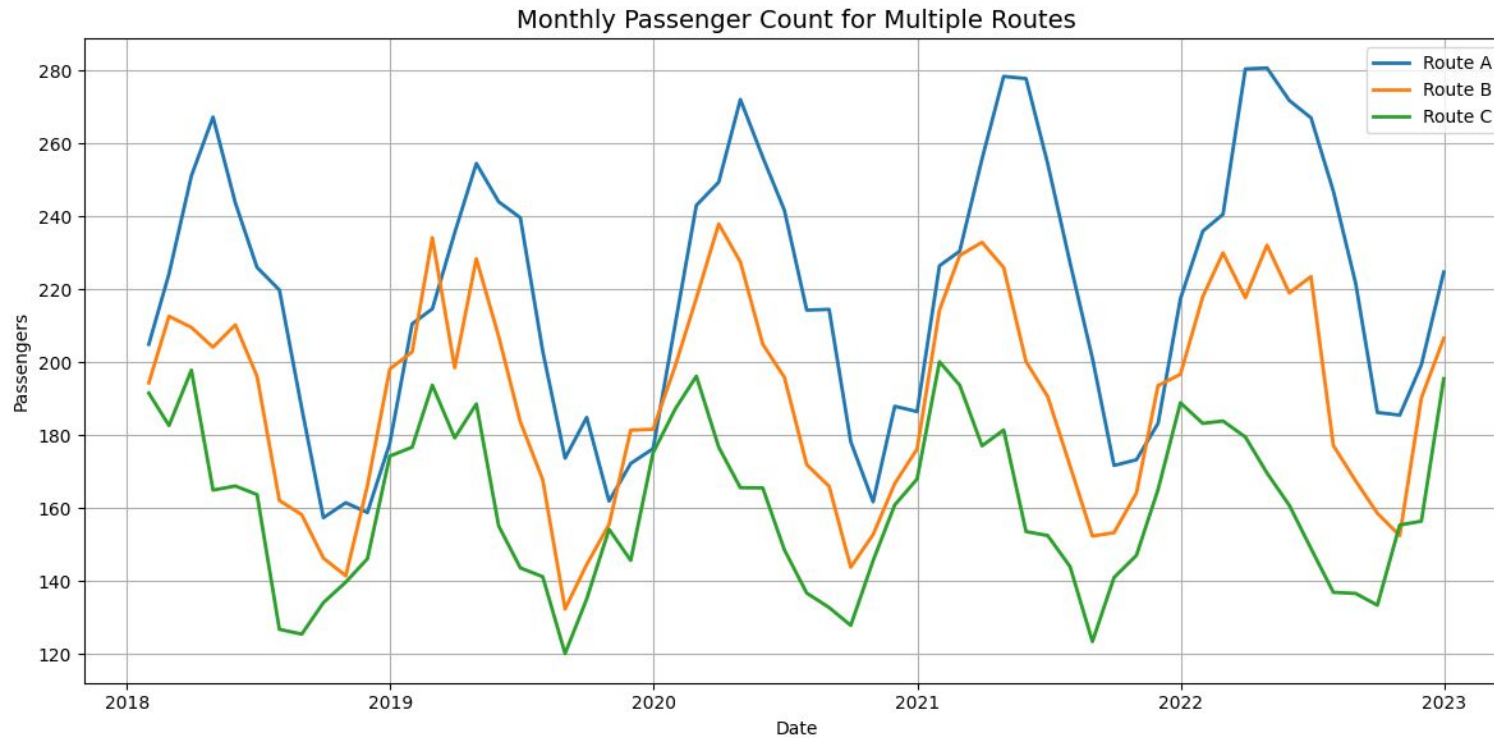


Inference:

- An upward trend (more passengers over time)
- Strong seasonality (recurring annual peaks and troughs)
- Increasing amplitude, which suggests multiplicative seasonality (seasonal changes grow as the trend increases).

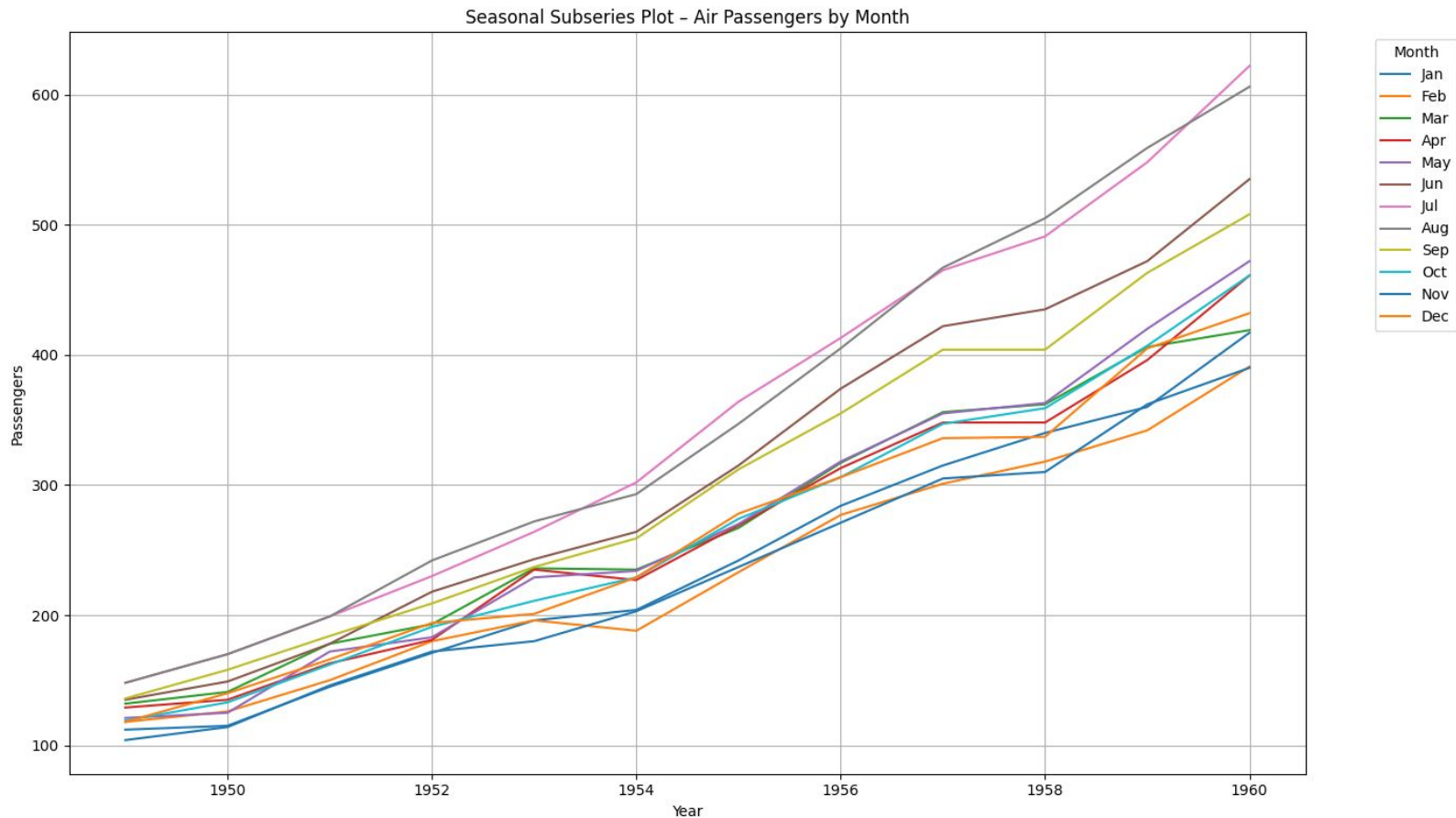
Multiple Line Plot (Shared Time Axis)

While all three routes display clear seasonal patterns, the timing of peaks and troughs varies across them



Seasonality (Recurring Patterns at Fixed Intervals)

Seasonal Subseries Plot



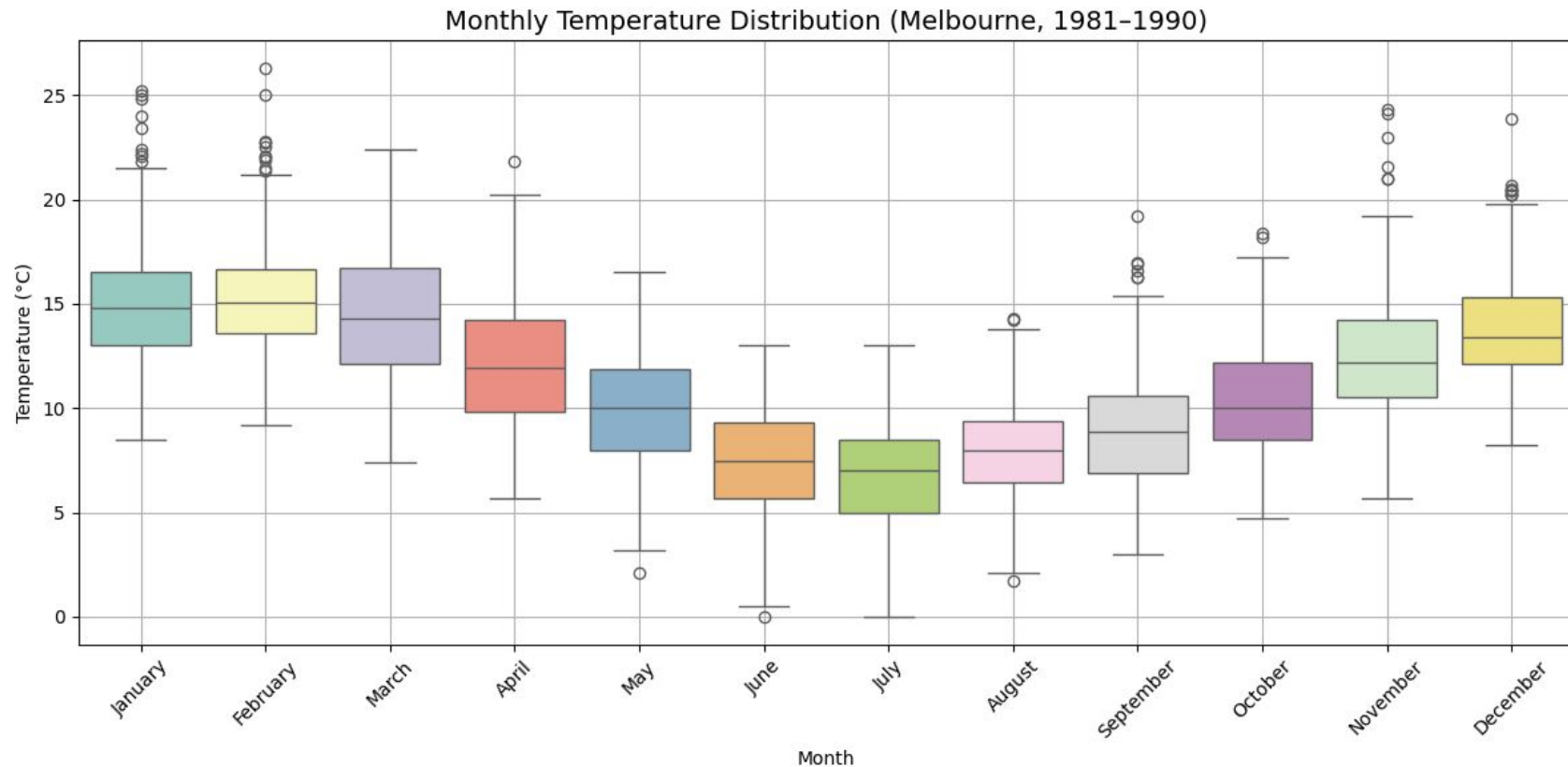
- July and August consistently show the highest passenger traffic—indicating a strong summer travel season.

Inferences

- Strongest months:
 - July and August consistently show the highest passenger traffic—indicating a strong summer travel season.
- Weaker months:
 - February and November tend to have the lowest passenger numbers, reflecting seasonal dips.
- Upward trend:
 - All months show a gradual increase in passengers, confirming a consistent long-term growth trend.
- Stable seasonal structure:
 - The relative positioning of months is preserved year over year, suggesting that seasonal behavior is consistent (no major shift in seasonal demand).

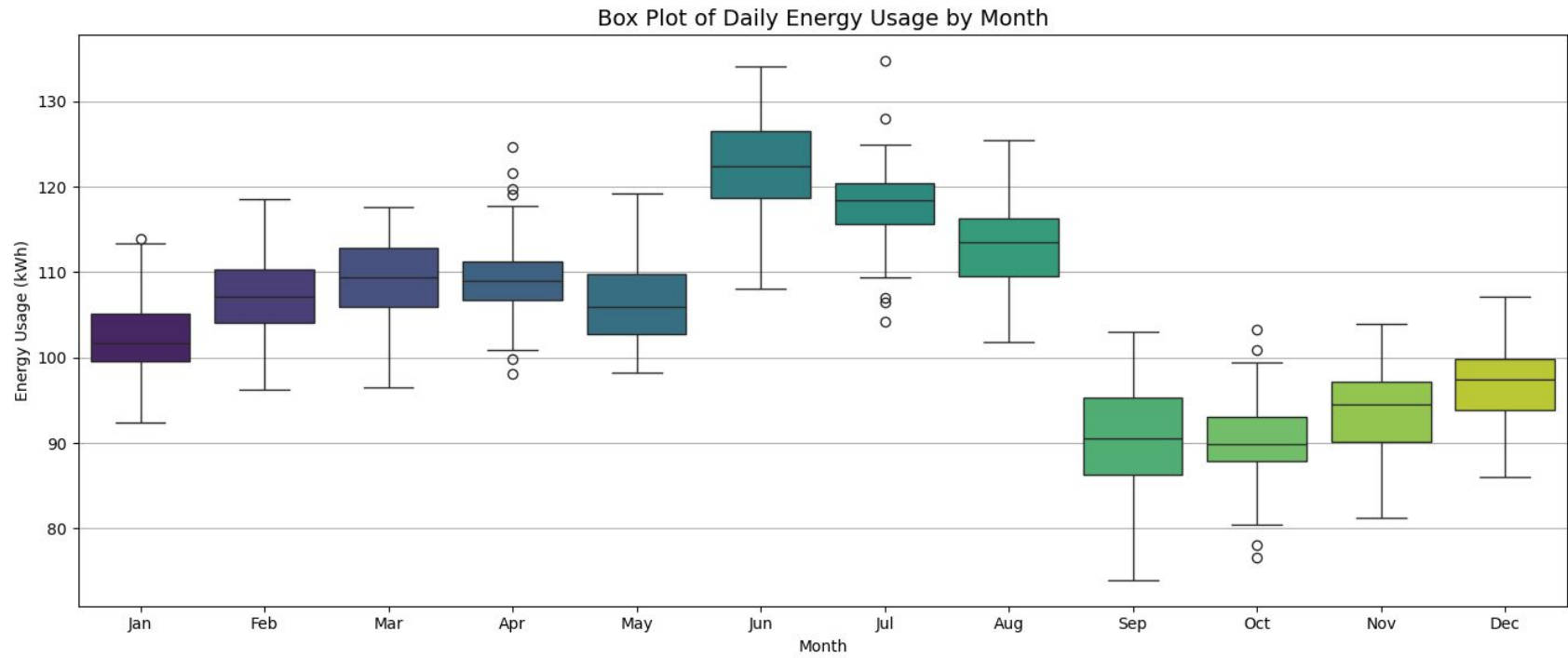
Box Plot by Time Unit

Box plots by time unit (such as by month or day of the week) are useful for visualizing the distribution, spread, and seasonal variations in a time series



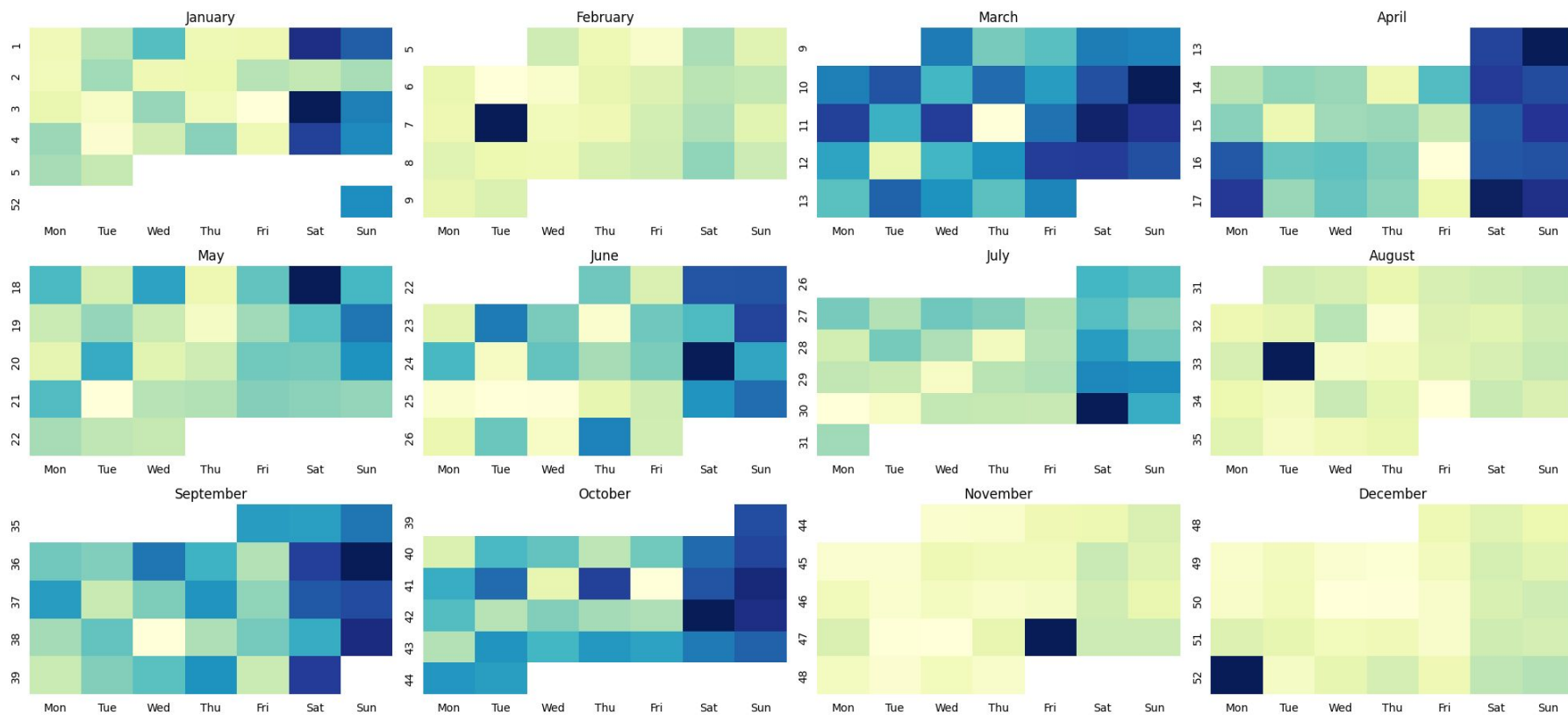
- **Coldest months:** June, July, and August (Australian winter).
- **Warmest months:** January, February, and December (Australian summer).

Box plot by month



Calendar plot

Calendar-Style Monthly Heatmaps of Daily Retail Sales (2023)

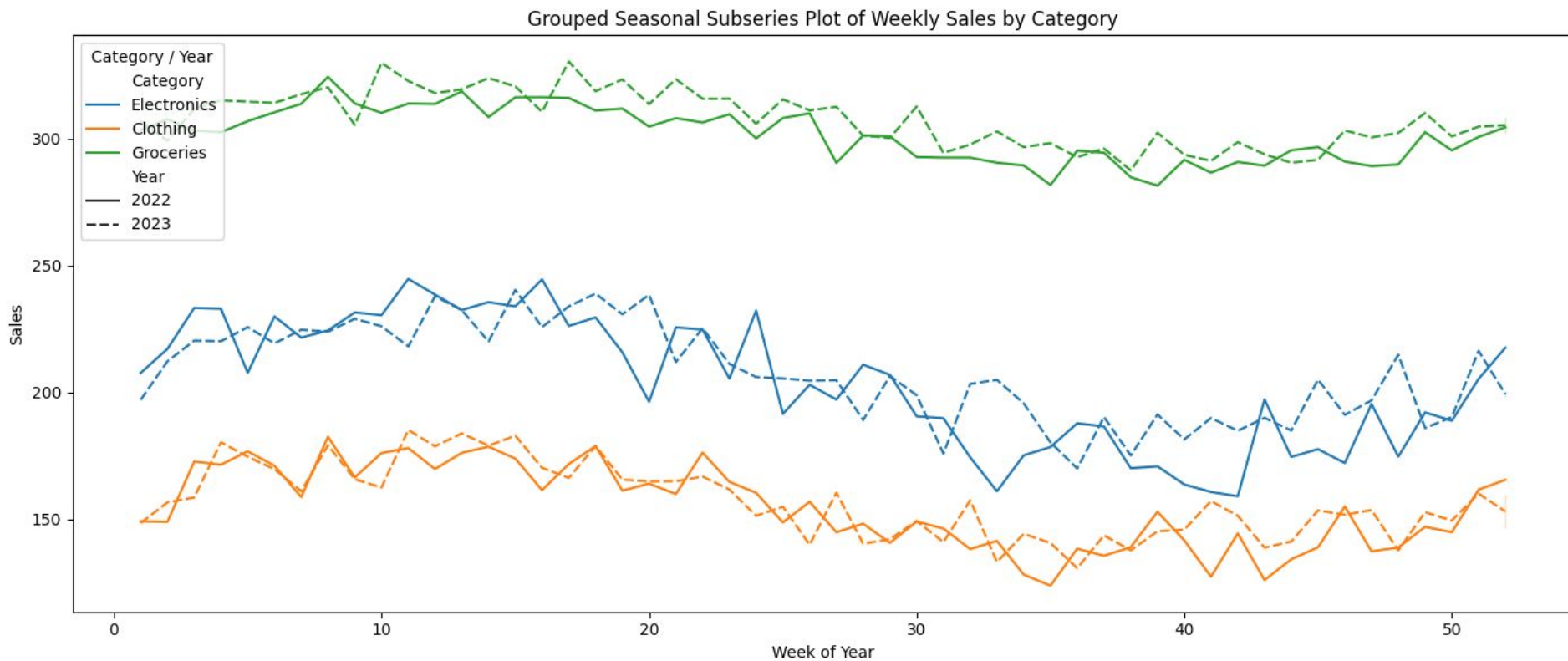


Calendar plot

- Inferences
- **Higher sales on weekends** are visible as consistently darker shades for Saturdays and Sundays across months.
- **Notable spikes** around February 14 (Valentine's Day), August 15 (Independence Day), November 24 (Black Friday), and December 25 (Christmas).
- Certain months like November and December show generally **increased activity**, aligning with holiday shopping behavior.

Grouped sub seasonal plots

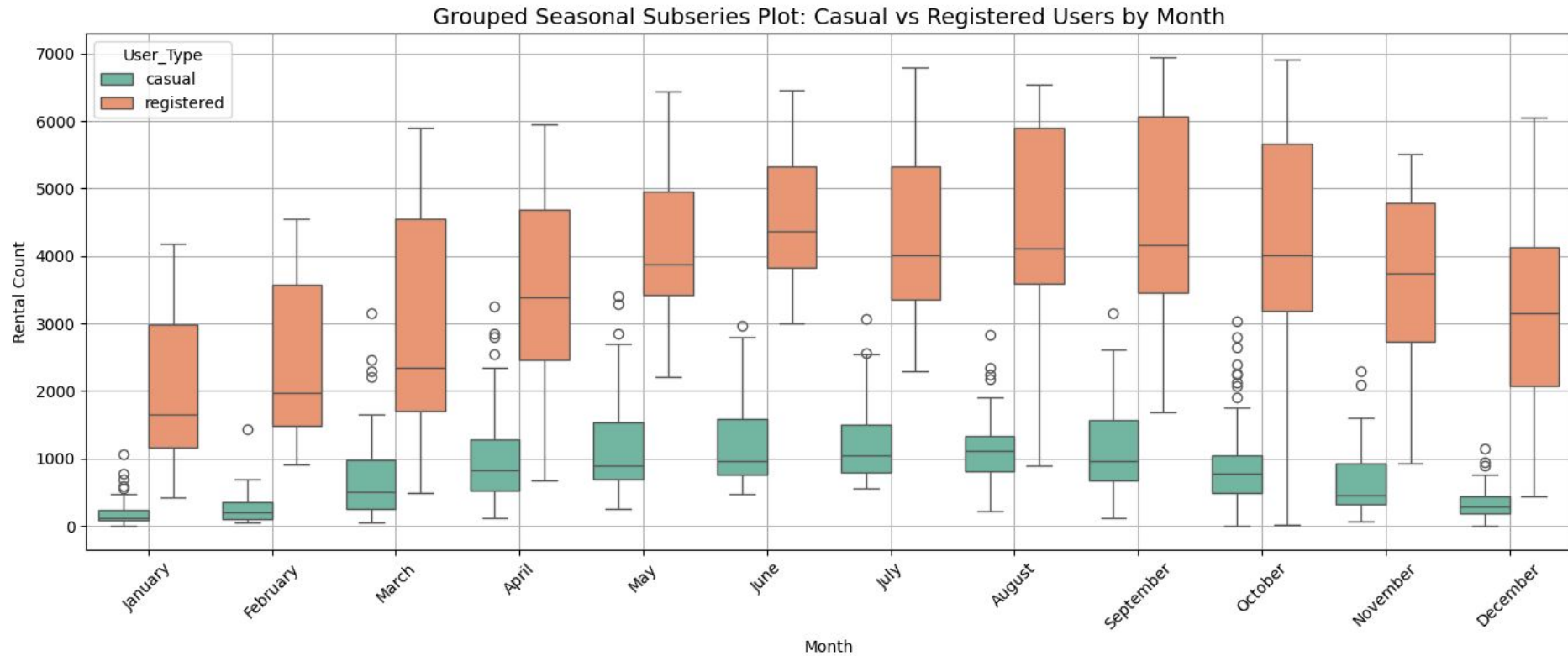
- Use Case: A business analyst at a retail chain wants to compare **seasonal trends in sales** across different product categories (e.g., Electronics, Clothing, Groceries) over the last two years



Inferences

- **Electronics** shows prominent peaks near year-end in both years, indicating strong seasonality around holiday seasons.
- **Clothing** exhibits gentler seasonal variation, but still follows noticeable cycles.
- **Groceries** show minimal seasonal variation but remain consistently high — essential purchases are less impacted by season.

Grouped sub seasonal plots – box plots



Grouped sub seasonal plots- box plots

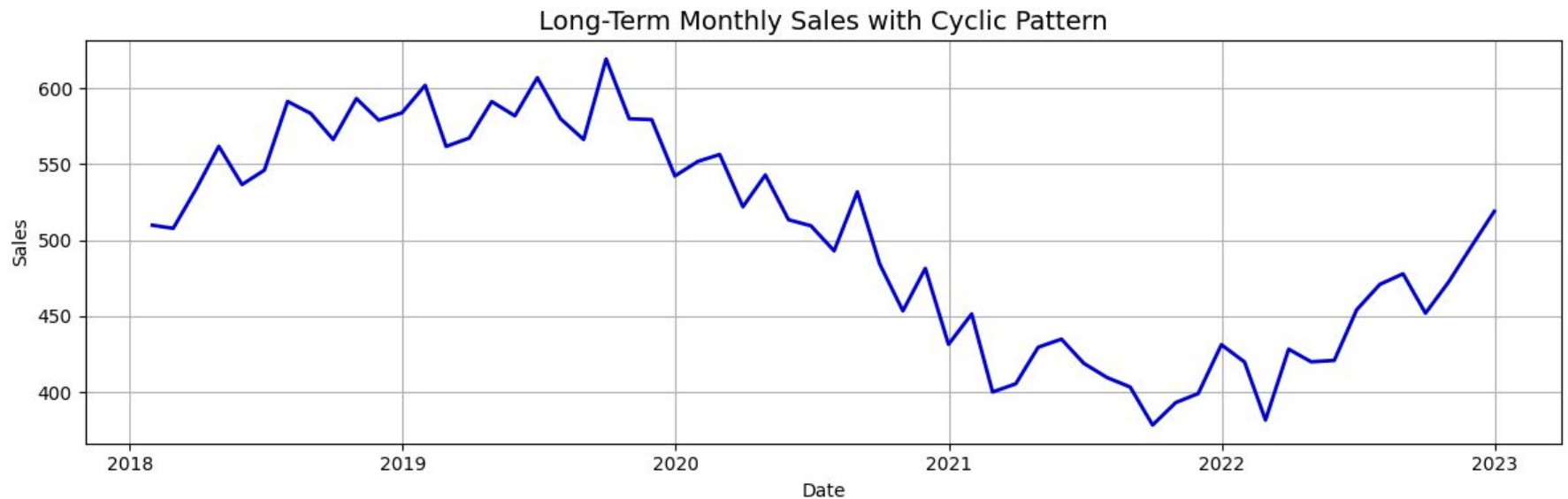
- Inferences

- Across all months, registered users consistently show much higher rental counts than casual users.
- Clear Seasonality for Both User Types
- Rental activity increases from March through October, peaking in July–September, then drops in November–February.
- Indicates a strong seasonal pattern, likely driven by weather or holidays.

Cyclicity (Long-Term Non-Fixed Repeating Patterns)

Cyclic Pattern

- Line plots over extended time frames can reveal **cyclic behavior** — slow, undulating trends that don't occur at regular intervals like seasonality, but still exhibit repeated patterns

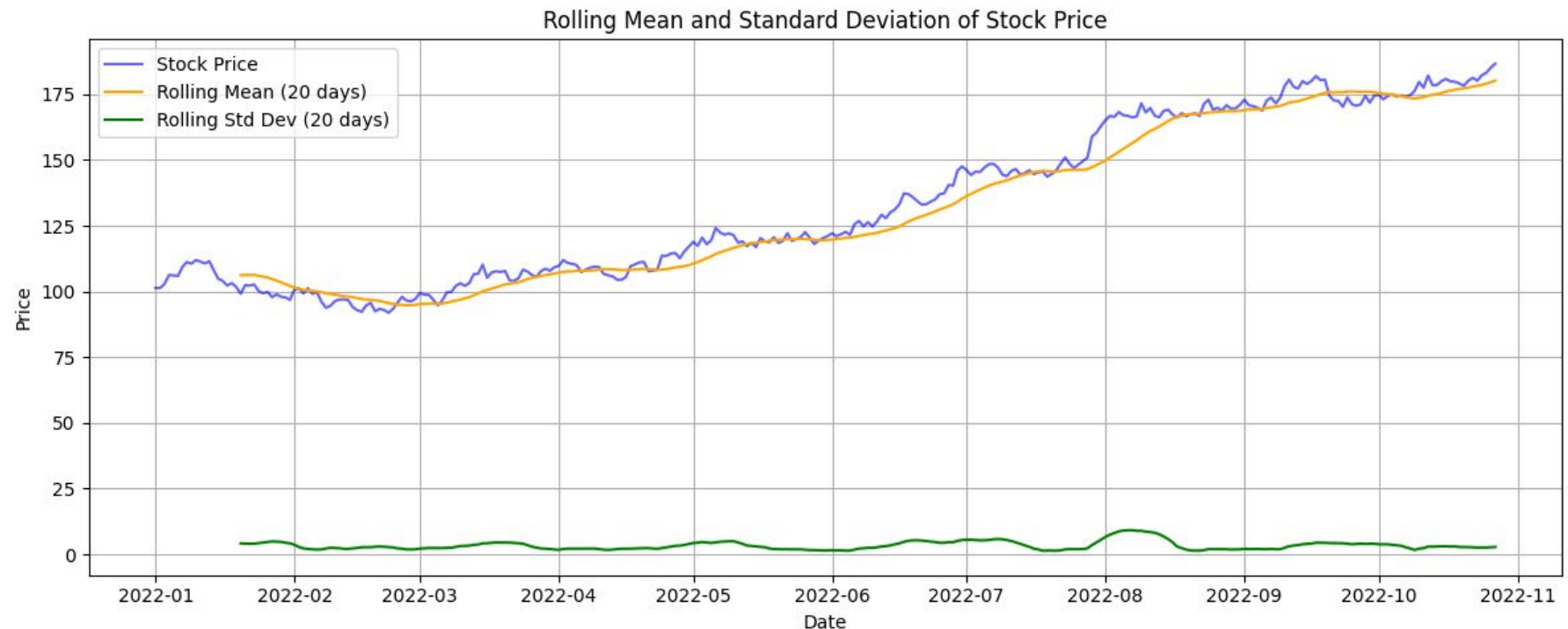


Stationarity

Rolling Mean and Standard Deviation Plot

- Rolling statistics help examine stationarity by evaluating how the mean and variance of a time series evolve over time.
- If they remain roughly constant, the series may be stationary.
- If they drift or change, the series likely violates the stationarity assumption required for many models.

Rolling Mean and Standard Deviation Plot



- The stock price clearly trends upward over time, which violates the constant mean assumption.
- The rolling mean increases steadily, reinforcing the presence of a non-stationary trend.
- The rolling standard deviation shows fluctuations, indicating changing volatility over time.
- This plot suggests that the series is non-stationary and may require differencing or transformation before applying forecasting models.

Stationarity 2: Differenced Series Plot

- Differencing a time series helps eliminate trends and stabilize the mean, which is a crucial step in transforming a non-stationary series into a stationary one.
- Plotting the differenced series allows visual inspection to see if the transformed series appears more stationary.

Use Case:

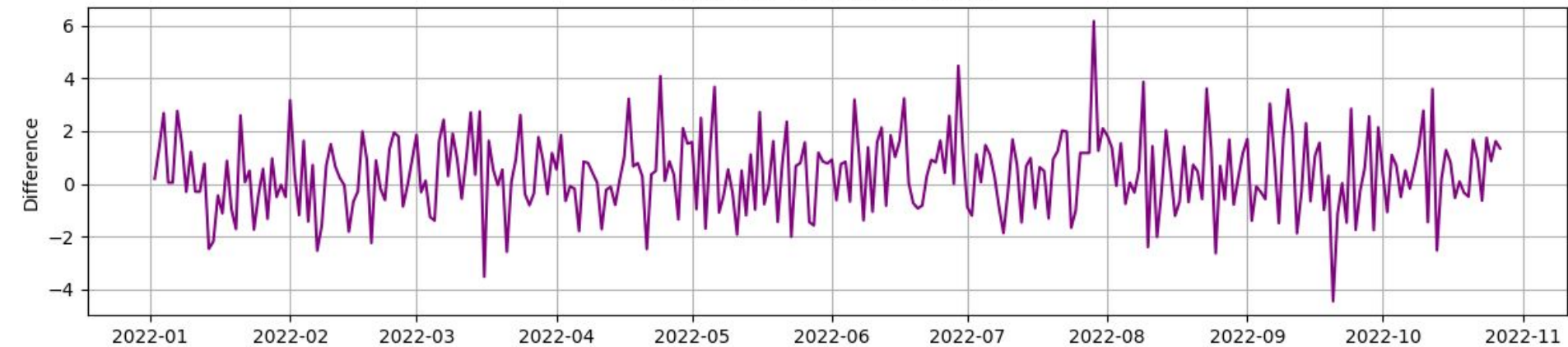
A financial data scientist analyzing a company's daily closing stock prices wants to check whether applying first-order differencing makes the series suitable for forecasting models like ARIMA.

Differenced Series Plot

Original Stock Price Series



First-Order Differenced Series



• Inference

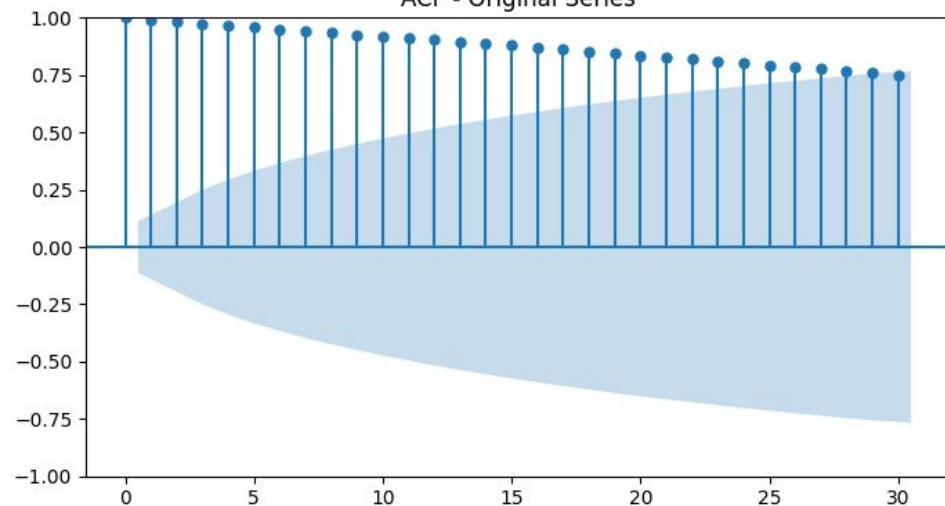
- The original series shows a clear upward trend, confirming non-stationarity.
- After first-order differencing, the series fluctuates around a constant mean, visually appearing more stationary.
- The transformed series is better suited for forecasting models that require stationarity, such as ARIMA.
- Additional statistical tests (e.g., Augmented Dickey-Fuller) can be used to confirm stationarity after differencing.

ACF and PACF Plots (Original and Differenced Series)

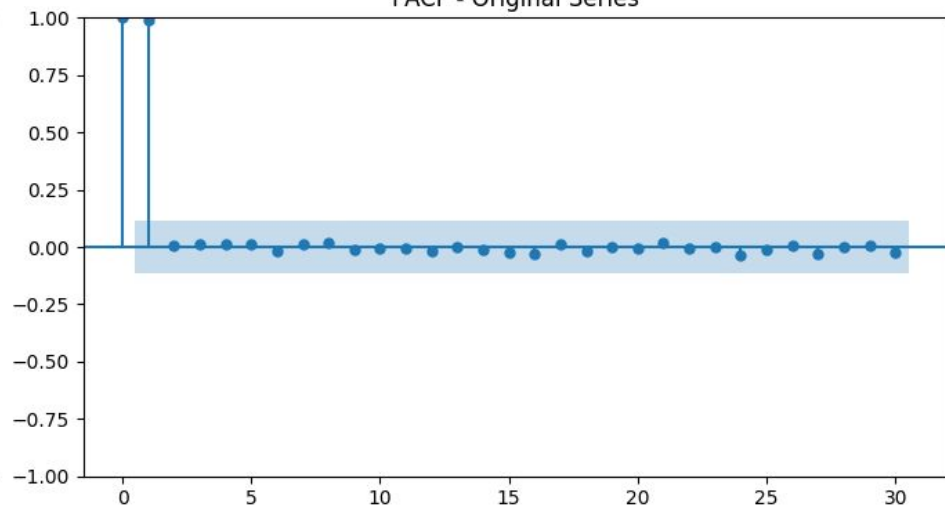
- Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots are used to examine the internal dependence structure of a time series.
- They are crucial for diagnosing stationarity and for identifying suitable lags in ARIMA-type models.
 - ACF shows correlation with lagged versions of the series.
 - PACF isolates the correlation at a specific lag, removing indirect effects
- Comparing ACF/PACF plots before and after differencing helps verify whether the transformation has successfully removed trend or seasonality.



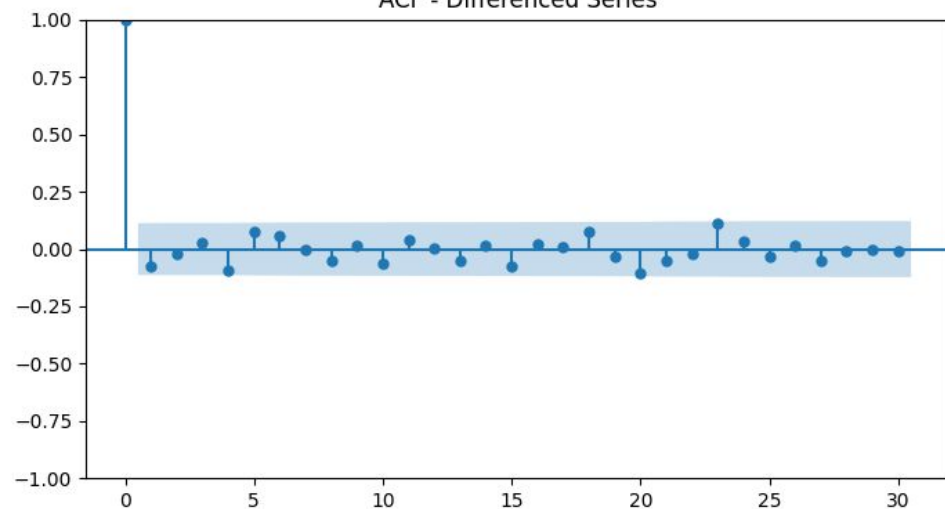
ACF - Original Series



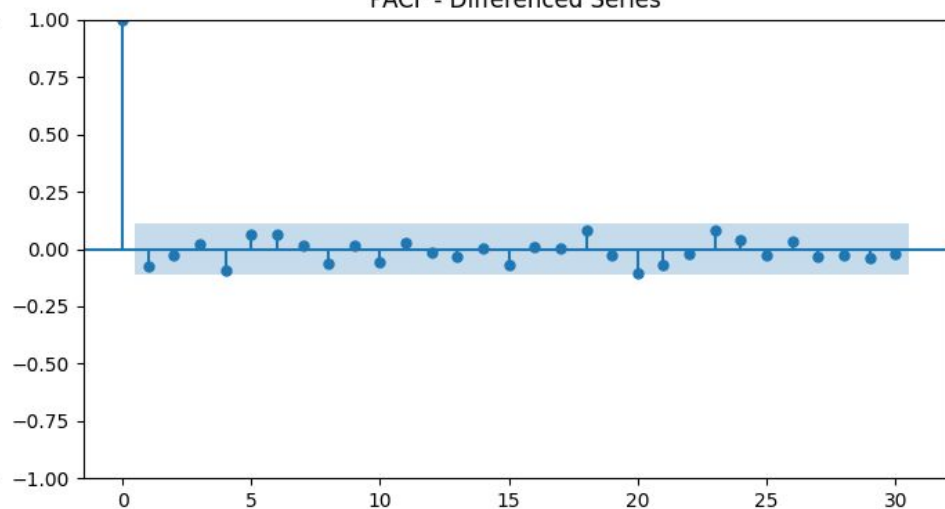
PACF - Original Series



ACF - Differenced Series



PACF - Differenced Series

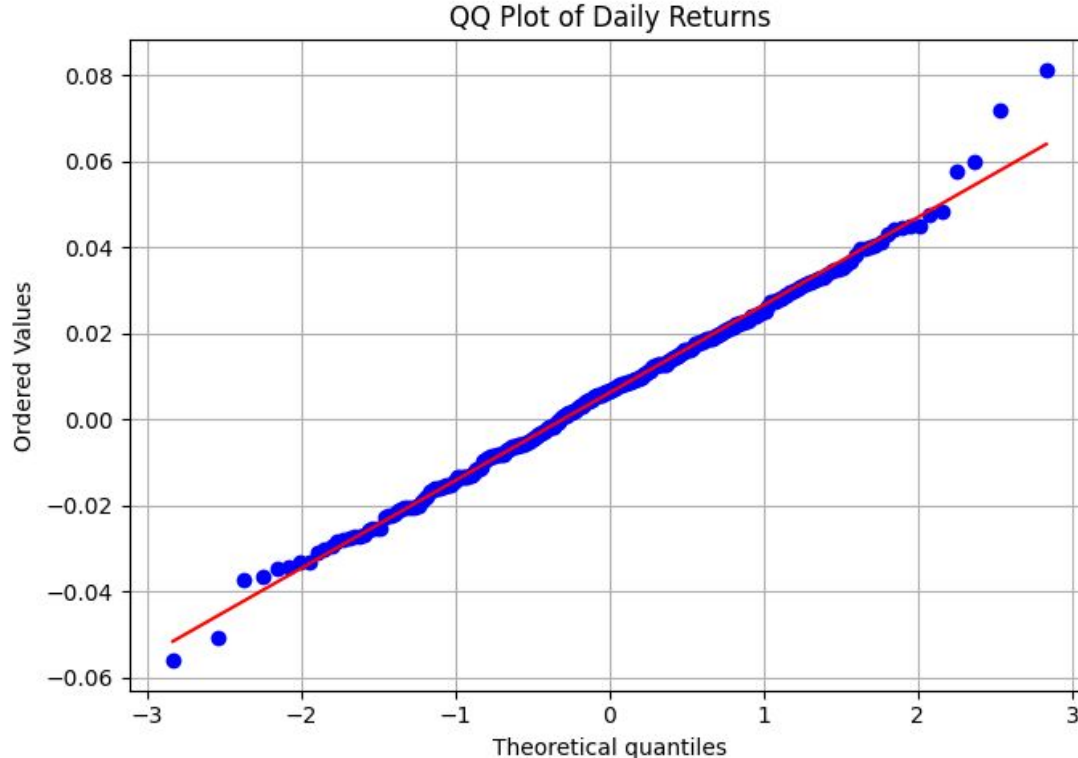


Inference

- Inference

- The original series shows high autocorrelation across many lags in the ACF plot, typical of a non-stationary series.
- PACF of the original also shows significant lags, suggesting trend or persistence in the series.
- After differencing:
 - The ACF drops off more rapidly — an indication of reduced correlation over time.
 - The PACF becomes more compact, suggesting that fewer lags are needed in the model.
- These results support the use of first-order differencing and guide the selection of p , d , q for ARIMA.

QQ Plot (Quantile-Quantile Plot)



A QQ (Quantile-Quantile) plot compares the distribution of a time series to a **normal distribution**.

- Inference

- The plot shows a noticeable deviation from the straight line, especially in the tails, indicating the presence of skewness and heavy tails.
- These distributional abnormalities suggest that the series may not satisfy the assumptions required for certain linear or Gaussian models.
- This insight is critical for selecting robust statistical models (e.g., GARCH, t-distributions) or transformations.

Autocorrelation (Self-Correlation Over Time)

ACF Plot (Correlogram)

- The Autocorrelation Function (ACF) plot shows how current values in a time series relate to past values (lags).

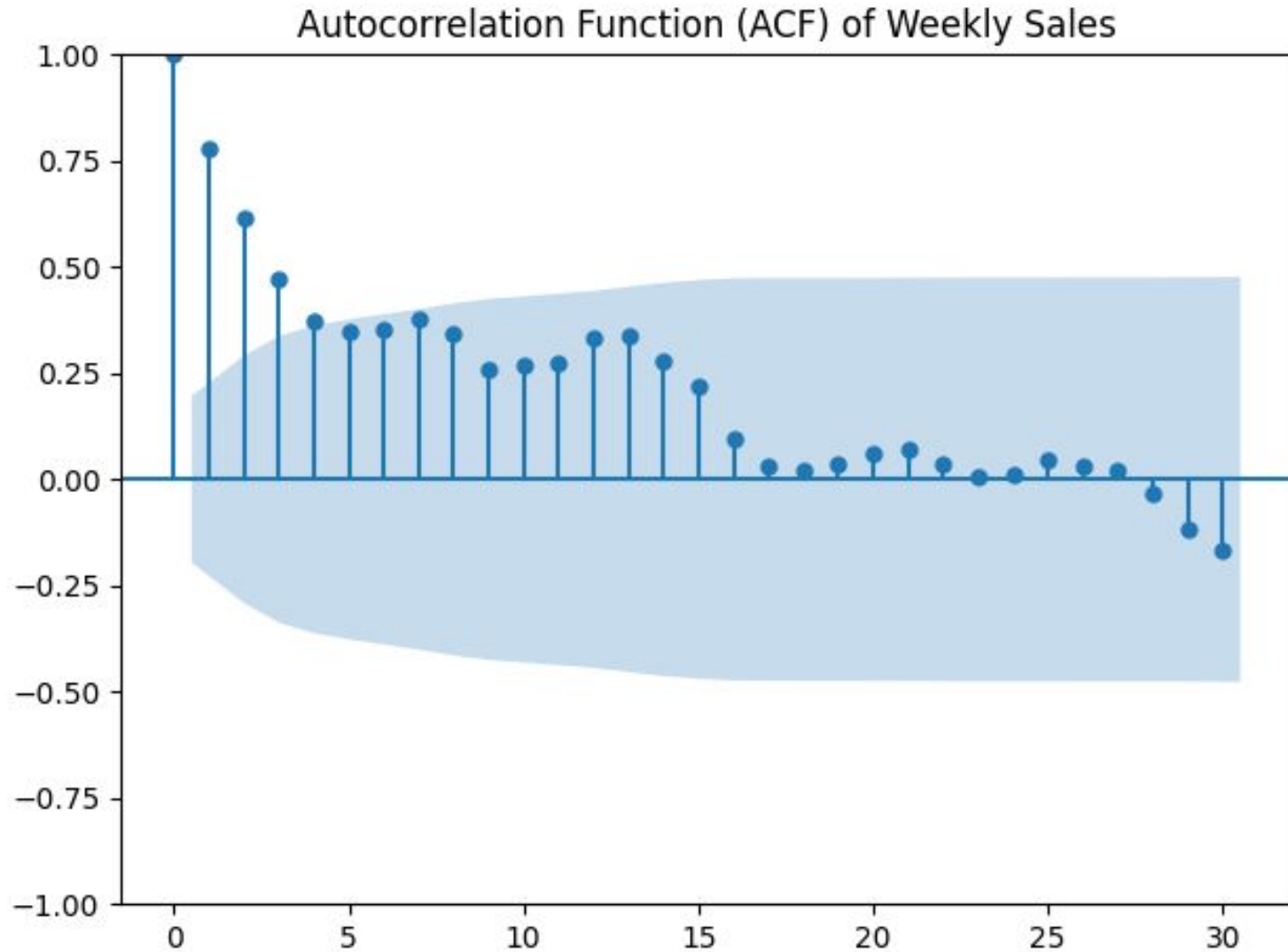
It's a foundational tool for:

- Detecting autocorrelation structure.
 - Determining how many past values influence the present.
 - Choosing lags for ARIMA models.
-
- Spikes at specific lags indicate strong linear dependence with previous time steps.

Use Case:

A demand forecasting analyst in retail wants to check how strongly the weekly sales of a product are correlated with past weeks to guide model selection for inventory management.

Auto Correlation Function



Auto Correlation Function

Inference

- The plot shows gradually declining autocorrelation, suggesting that past values strongly influence future values.
- This is typical of AR(1) processes or similar autoregressive models.
- The insight supports the use of autoregressive terms in time series forecasting models like ARIMA or SARIMA.

PACF Plot

The Partial Autocorrelation Function (PACF) plot shows the direct effect of a lagged value on the current value, removing the effects of intermediate lags.

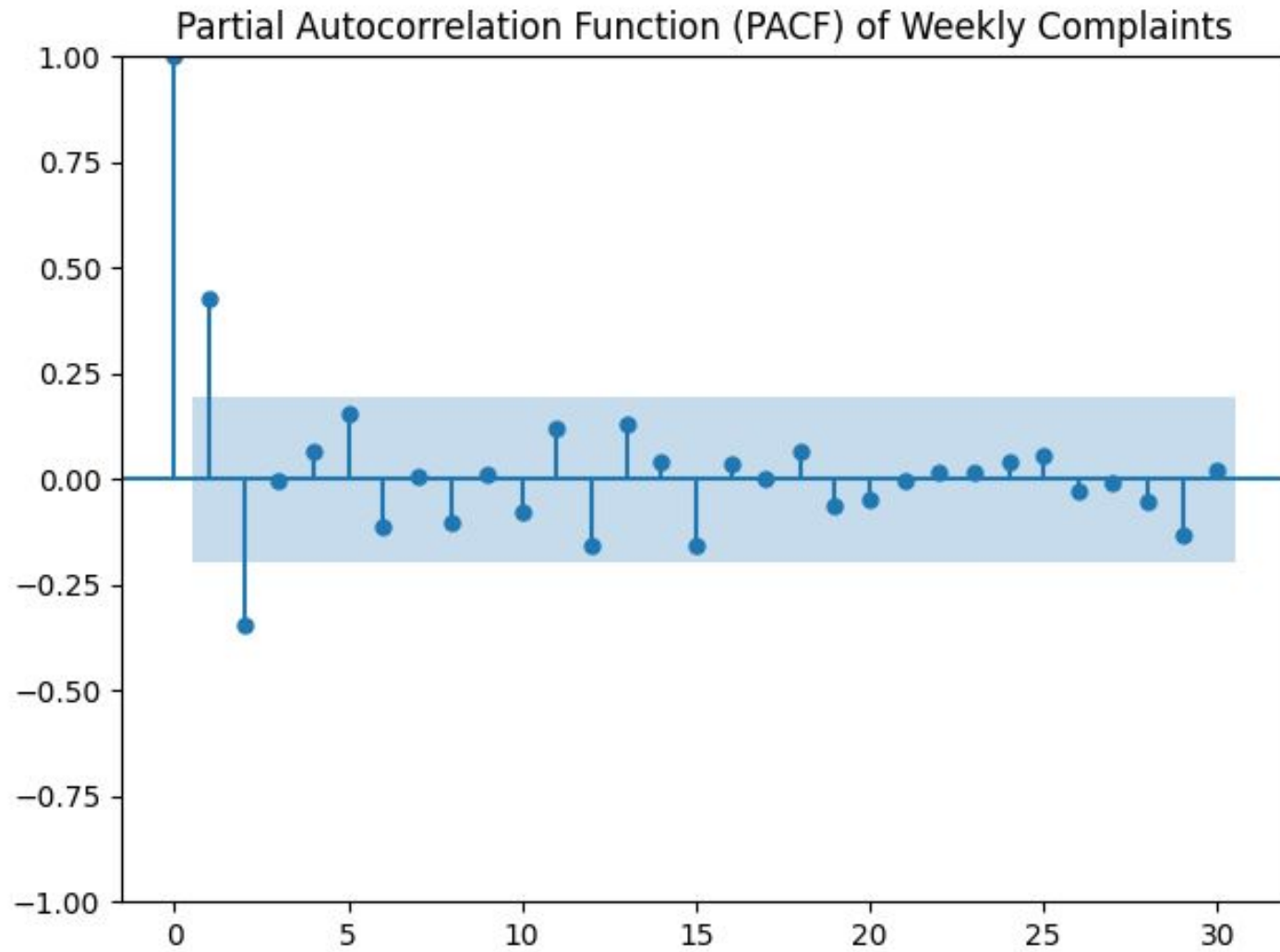
It helps determine:

- The appropriate lag order for autoregressive (AR) models.
- Whether dependencies are due to direct or indirect relationships.
- How many lags should be retained in model design.

Use Case:

- A telecom network operator analyzes weekly customer complaint counts to identify how far back complaints directly affect the current week's value — helping design predictive maintenance and resolution models.

PACF



- Inference

- The plot shows significant spikes at lag 1 and lag 2, then a sharp drop-off — suggesting that only the first two lags directly impact the current value.
- This supports the use of an AR(2) model.
- The PACF plot clarifies the true order of autoregression, reducing overfitting in models like ARIMA.

Lag Plot

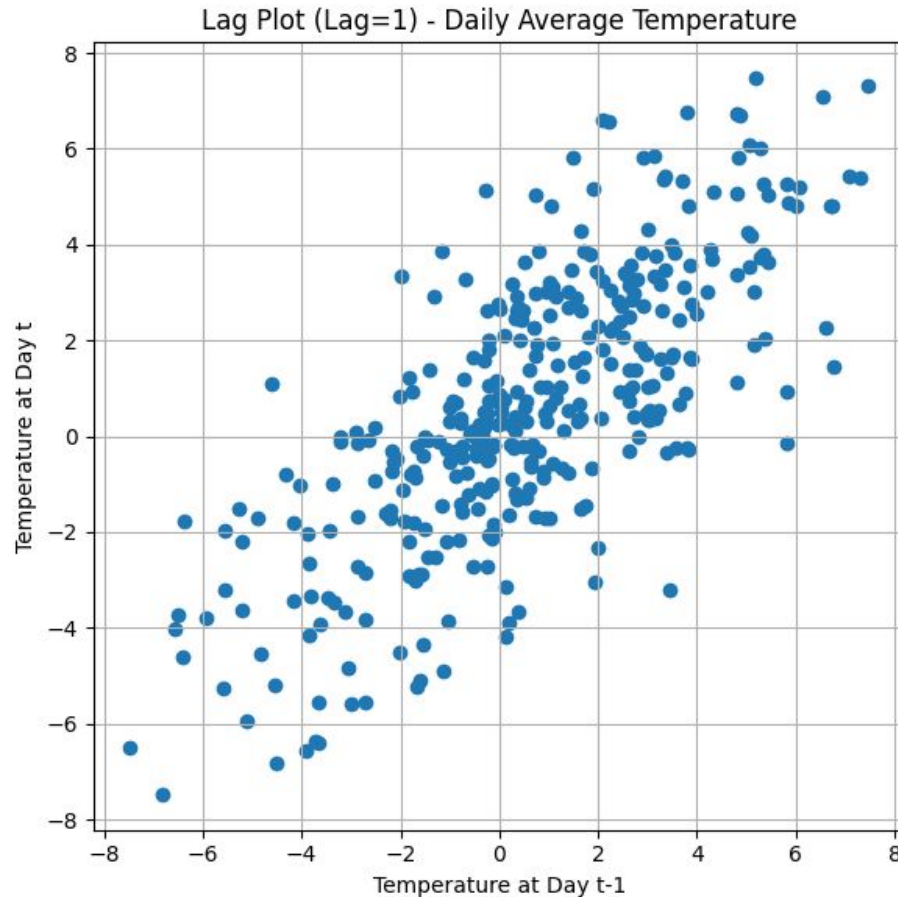
Lag plots visualize the relationship between a time series and a lagged version of itself, helping to identify:

- Serial correlation (i.e., dependence across time).
- Linearity or non-linearity in the relationship.
- Whether a time series is random or has structure.

Use Case:

A weather department analyzes daily average temperatures to see if today's temperature is correlated with previous days, which helps in short-term climate modeling.

Lag plot

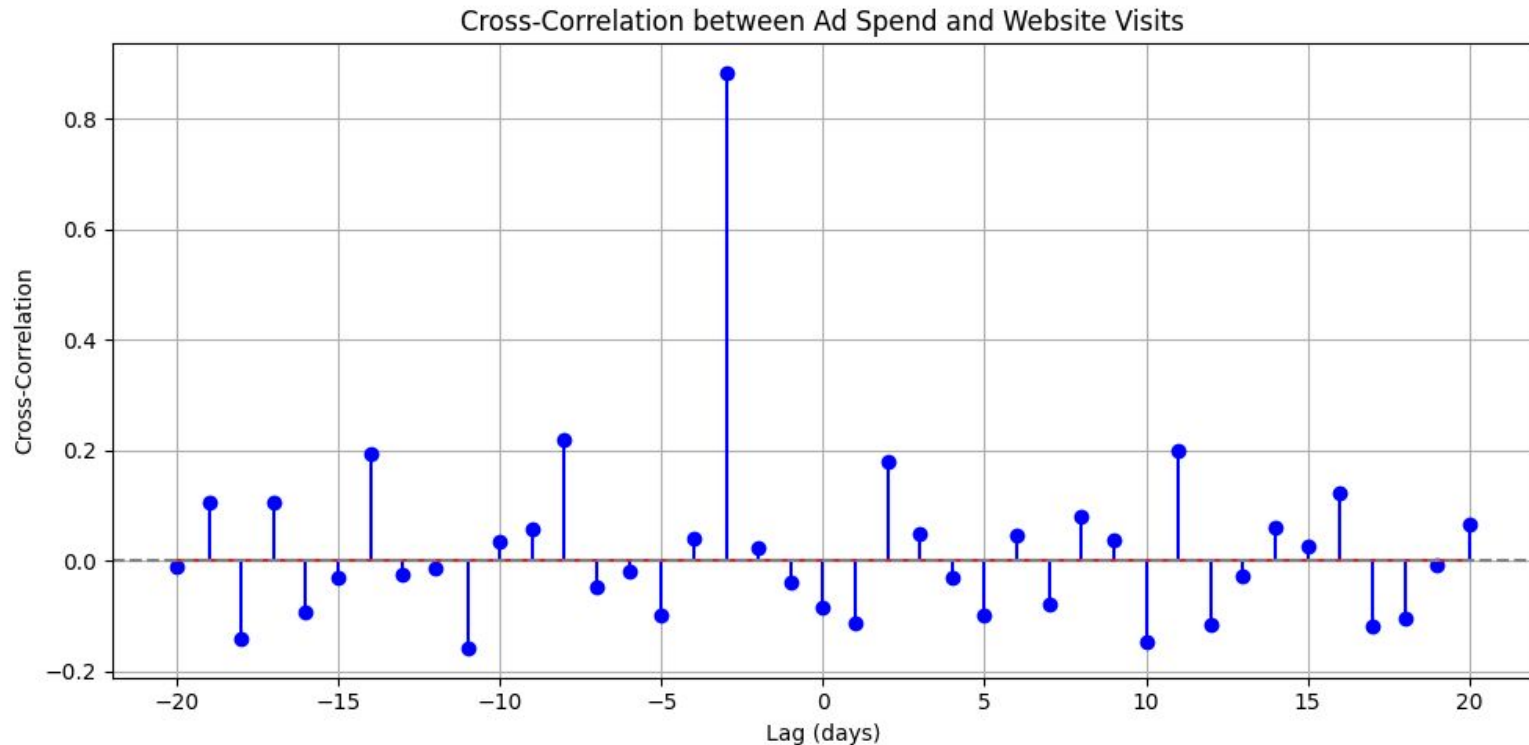


- The strong diagonal linear pattern indicates positive autocorrelation, meaning today's temperature is similar to yesterday's.
- The absence of a cloud-like spread suggests the series is not random.
- Lag plots are a quick visual diagnostic tool to detect autocorrelation before running statistical tests or models

Cross-Correlation Plot (Multivariate)

- Cross-correlation plots measure the correlation between two time series at different lags, helping uncover:
 - Lead-lag relationships (e.g., one variable leads or predicts another).
 - Delayed effects between variables.
- Temporal dependencies in multivariate systems.
- Use Case: A marketing team analyzes the lag between advertising spend and website visits to understand how long after a campaign traffic begins to increase.

Cross-Correlation Plot (Multivariate)

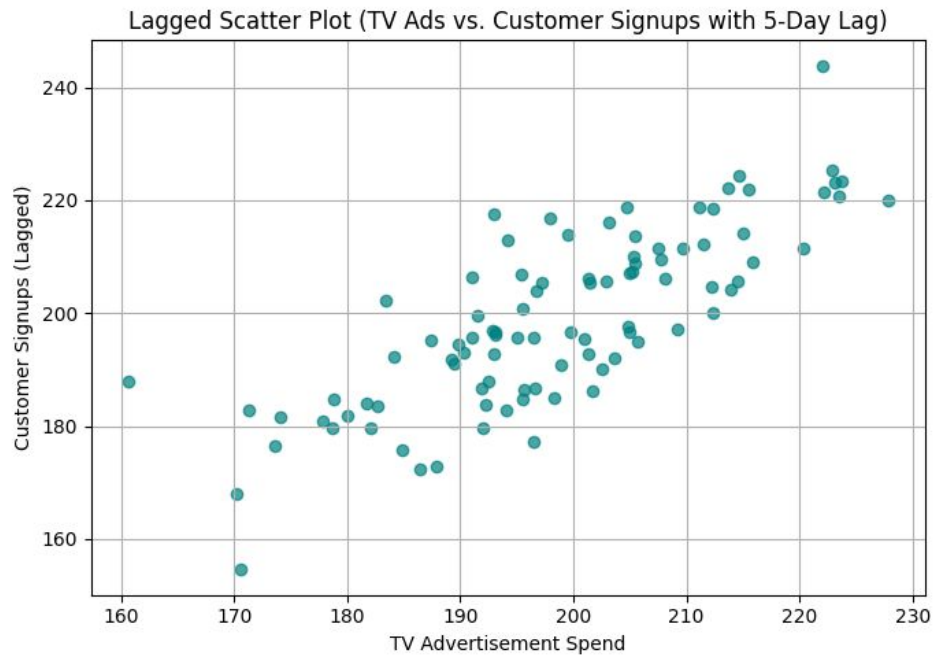


- Inference

- The highest correlation is observed at a positive lag of 3, indicating that advertising spend leads to increased website traffic with a 3-day delay.
- Negative lags have lower correlation, suggesting past visits don't drive ad spend.
- Cross-correlation plots are vital in causal analysis, campaign planning, and forecasting with external regressors.

Lagged Scatter Plot (Series A vs. Series B with Lag)

- Lagged scatter plots help visualize the relationship between two time series when one is shifted in time.
- This plot is especially helpful in identifying whether one variable leads or lags another, revealing causal or delayed influence between them.
- Use Case:
 - A telecom company wants to determine whether there's a delayed response between TV advertisement spend and new customer signups.
 - This insight can guide how long after a campaign to expect impact.



Inference

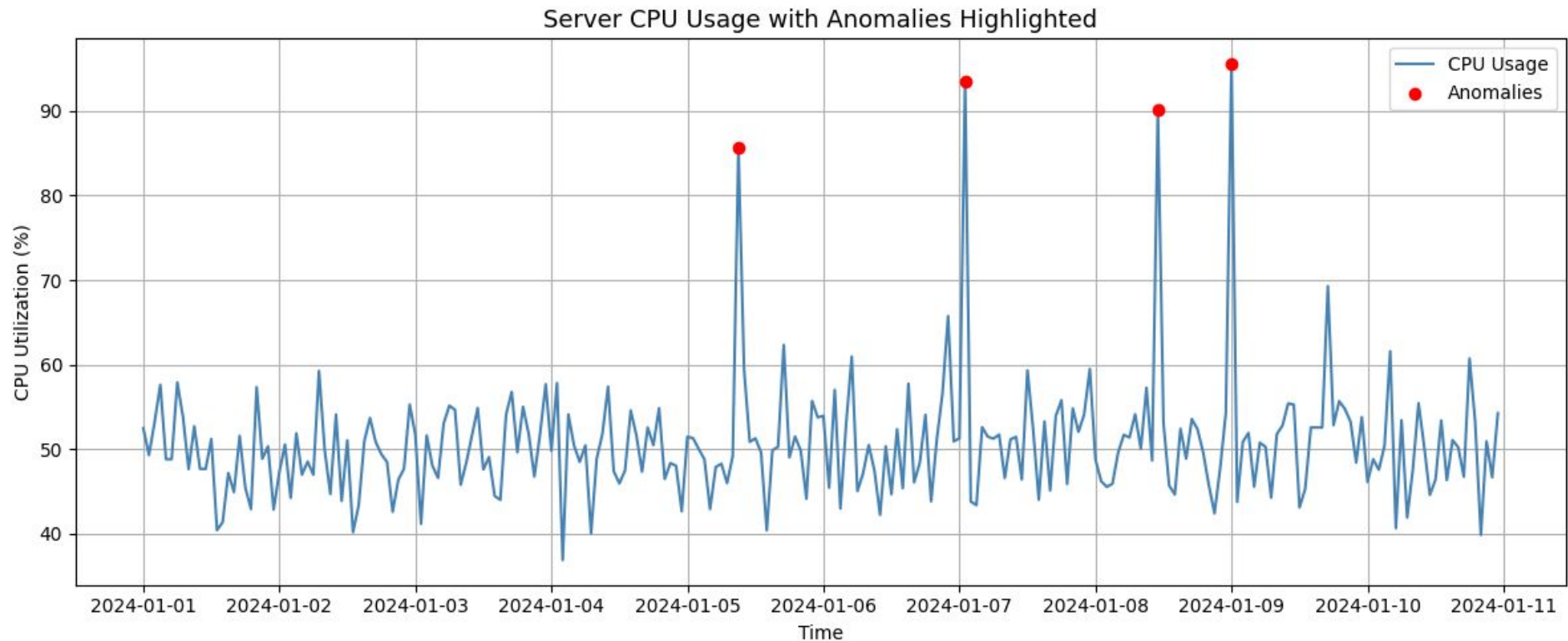
- The plot shows a strong positive correlation, indicating that higher ad spending leads to increased customer signups after 5 days.
- This confirms a lagged causal relationship, which is essential for marketing impact analysis.
- Helps the business plan campaign durations and expect realistic lead times between ad spend and customer acquisition

Outliers and Anomalies

Time Series Line Plot with Anomaly Overlay

- Line plots with anomaly overlays highlight unusual values that deviate significantly from expected patterns, helping in:
 - Detecting unexpected events, system failures, or fraudulent behavior.
 - Monitoring real-time systems like servers, IoT sensors, finance, or healthcare.
 - Visualizing anomalies inline with normal temporal trends.
- Use Case:
 - A server monitoring team tracks CPU utilization over time.
 - Spikes far above the average could signal system stress, intrusion attempts, or bugs in software deployments.

Time Series Line Plot with Anomaly Overlay



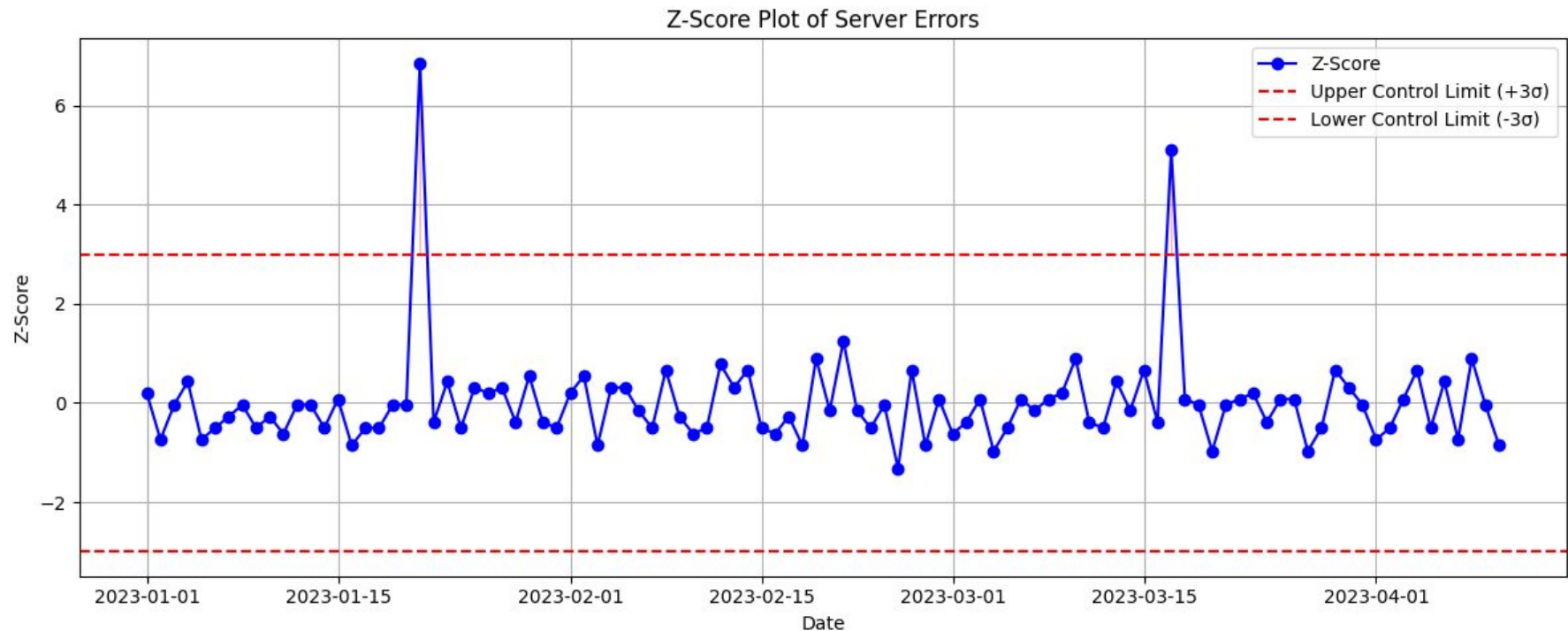
- Inference

- The red points indicate anomalous spikes in CPU usage, likely caused by unexpected loads or system events.
- Most of the time, usage stays between 40%–60%, suggesting a stable baseline.
- Time series anomaly overlays like this are vital for real-time monitoring dashboards and alerting systems.

Z-Score Plot / Control Chart

- A Z-score plot shows how many standard deviations a data point is from the mean. It's commonly used in control charts to monitor if a process is operating within expected limits.
- Useful for:
 - Identifying extreme values beyond thresholds (e.g., $\pm 3\sigma$).
 - Monitoring process stability or drift.
 - Flagging early signs of anomalies in production systems, websites, or operations.
- Use Case:
 - A website operations team tracks daily server error counts and uses a Z-score plot to detect when the error rate spikes unusually high, potentially due to code deployments or cyberattacks.

Z-Score Plot / Control Chart



- Inference

- Two spikes cross the **+3σ threshold**, indicating **statistical anomalies** in server errors.
- These points likely correspond to **unexpected operational issues**.
- The rest of the data falls within control limits, suggesting **stable behavior**.
- Z-score plots are highly effective for **flagging abnormal values** in real-time monitoring systems.

Q&A